

Masoud Kargar

Senior .NET Backend Developer | AI Engineer

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PROFESSIONAL SUMMARY

Senior .NET Backend Developer and AI Engineer with 6+ years of experience building scalable backend systems, REST APIs, microservices, and data-intensive applications using C# and .NET. Strong background in Clean Architecture, CQRS, SQL optimization, distributed authentication, and message-driven systems. Improved production API/SQL performance by approximately 35% through query optimization and architectural refinement. Hands-on experience building AI-powered applications with LLMs, Retrieval-Augmented Generation (RAG), embeddings, semantic search, ChromaDB, OpenAI-compatible APIs, OpenRouter, and local Persian-language embedding services. Experienced in integrating AI services with production backend architectures and designing maintainable, provider-independent systems.

CORE AI ENGINEERING

- LLM & Generative AI: LLM application development, prompt engineering, system prompts, OpenAI-compatible APIs, OpenAI, OpenRouter, streaming and non-streaming generation, provider abstraction
- RAG & Knowledge Systems: Retrieval-Augmented Generation, document ingestion, chunking and retrieval pipelines, embeddings, semantic search, context augmentation, metadata-aware storage, RAG question answering
- Embeddings & Vector Search: OpenAI embeddings, local embeddings, Persian/Farsi embeddings, vector similarity search, ChromaDB, persistent vector storage
- AI Infrastructure: Python, FastAPI, Uvicorn, local embedding microservices, AI/LLM API design, Docker-based services, separation of embedding and chat providers, backend integration and deployment

CORE BACKEND ENGINEERING

- Languages & Frameworks: C#, .NET 8, ASP.NET Core, TypeScript, JavaScript, React, Blazor WebAssembly, .NET MAUI
- Architecture: Clean Architecture, CQRS, Microservices, RESTful API Design, scalable service design, multi-tenant systems
- Databases & Caching: SQL Server, T-SQL, PostgreSQL, MongoDB, MySQL, Redis, Entity Framework, Dapper
- Messaging & Scheduling: RabbitMQ, Apache Kafka, Quartz
- Security & Identity: Keycloak, IdentityServer/Duende, JWT, OAuth2, OpenID Connect (OIDC), CORS
- DevOps & Infrastructure: Docker, Kubernetes, Nginx, Linux, ELK Stack, Git
- Other: Web3.js, blockchain application development

PROFESSIONAL EXPERIENCE

Senior .NET Backend Developer | *Rasta Novin Aria* | *Tehran, Iran* | *Apr 2025 – May 2026*

- Designed and developed backend services and RESTful APIs using .NET 8 as part of the core engineering team.
- Improved production SQL query and API performance by approximately 35% through targeted query optimization and architectural refinements.
- Implemented distributed authentication and authorization strategies using Keycloak and JWT across services.

Senior Backend Developer | *Setigh Fara Ofogh* | *Tehran, Iran* | *Oct 2024 – Mar 2025*

- Applied Clean Architecture principles to improve maintainability, testability, and long-term scalability.
- Led migration of a monolithic application toward a microservices-based architecture, improving deployment flexibility and resilience.
- Implemented Keycloak-based authentication for secure inter-service and user-facing communication.

Senior Backend Developer — Freelance | *Self-Employed* | *Pakdasht, Iran* | *Jul 2023 – Oct 2024*

- Built multi-tenant APIs enabling multiple client organizations to operate securely on shared infrastructure.
- Developed blockchain-based applications for client projects using Web3 technologies.
- Designed and implemented scalable backend systems following modern architecture and engineering practices.

.NET Backend Developer | *Behineh Kavan Keyfiat* | *Tehran, Iran* | *Jul 2021 – Jul 2023*

- Contributed to the design, development, and maintenance of backend services and RESTful APIs using .NET and SQL Server.

Back-End Developer — Freelance | *Self-Employed* | *Tehran, Iran* | *Mar 2020 – Jul 2021*

- Delivered backend development services for client projects using .NET and SQL Server.

SELECTED AI PROJECT

RAG API — Retrieval-Augmented Generation System | *GitHub: <https://github.com/MassoudKargar/RAG>*

- Built an OpenAI-compatible RAG API integrating document embeddings, ChromaDB vector storage, semantic search, and LLM chat generation.
- Implemented support for OpenRouter and OpenAI-compatible chat providers while keeping the embedding layer independently replaceable.
- Built a separate local embedding microservice using FastAPI/Uvicorn and the Persian embedding model *xmanii/maux-gte-persian*.
- Implemented vector operations for collection initialization, document ingestion, and semantic document search, plus streaming and non-streaming chat responses.
- Designed the architecture so the RAG service can act as a drop-in OpenAI API replacement with retrieval capabilities.

EDUCATION

- Diploma

CERTIFICATIONS & TRAINING

- Machine Learning with Python — Maktabkhooneh (2026)
- Microservice Architecture — Nikamooz, 70 hours (2023)
- PRO .NET Ecosystem — Nikamooz, 110 hours (2022)
- ASP.NET MVC — Cando Academy (2019)
- Programming Fundamentals — Cando Academy (2019)

LANGUAGES

- Persian — Native
- English — Upper Intermediate

ADDITIONAL

- Military Service — Completed